THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** A is a better conductor	Faster fall means faster conduction.
2	**(a)** reversed	Land breeze.
3	**(d)** sea breeze	Fluid motion.
4	**(d)** it absorbs more radiation	Dark absorbs.
5	**(d)** nearly stops	Temperatures equalise.
6	**(b)** heating iron rod	Solid transfer.
7	**(a)** hot air rising pushes it	Convection currents.
8	**(c)** not form	No temperature difference.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R describes the result of the flow from hot to cold, supporting A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Only radiation can travel through empty space, so R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heating from the top gives little convection (1); From the bottom, hot liquid rises and cooler sinks (1).	1+1
12	Evaporation takes heat (1); Keeps water cool (1).	1+1
13	Blanket is an insulator (1); Slows heat entering (1).	1+1
14	Metal conducts heat to the fins (1); Black radiates heat well (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thermocol best insulator (1); Ceramic next (1); Metal loses most heat (1).	1+1+1
16	Steel conducts (1); Plastic does not (1); Steel handle becomes hot (1).	1+1+1
17	Trap air (1); Insulation (1); Body heat is retained (1).	1+1+1
	OR: Hot air rises above (1); Small windows reduce radiation (1); Thick walls slow conduction (1).	1+1+1
18	Coolant absorbs heat inside (1); Releases it outside (1); Back coils warm (1).	1+1+1
	OR: Metal handle warms quickly (1); Wooden handle remains cooler (1); Conductivity differs (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Conduction (1). (ii) Wood is a poor conductor (insulator) of heat (1). (iii) Convection (1). (iv) Water at the bottom gets heated, becomes lighter and rises (1); cooler, heavier water from the top moves down and a convection current heats all the water (1).	1+1+1+2
20	Reflector concentrates light (1); Black inner surface absorbs (1); Glass lid traps heat (1); Insulated box reduces loss (1); Food cooks slowly (1).	1+1+1+1+1
	OR: Wool traps air (1); Cotton lets air pass (1); Insulated lunch boxes (1); Fridges (1); Reflective foil (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** heats and cools quickly	Low heat capacity.
2	**(a)** hot gases rise and draw fresh air	Draught by convection.
3	**(a)** white reflects most radiation	Less absorption.
4	**(b)** water cools slowly	Slow cooling.
5	**(b)** hot air can escape	Hot air rises.
6	**(c)** metal conducts heat away from the body faster	Rate of heat loss.
7	**(d)** they reduce loss of radiation	Trap heat.
8	**(c)** higher than near the floor	Hot air rises.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but one does not explain the other.
10	**(c)** A is true but R is false.	Land heats faster, warm air rises over land and cool air comes in from the sea, so A is true; R is false because land, not sea, becomes hotter.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Black surface absorbs heat from flame (1); Improves heating (1).	1+1
12	Glass conducts heat better (1); Takes heat from the hand faster (1).	1+1
13	Sand heats and cools quickly (1); Clear sky lets heat radiate out (1).	1+1
14	Reflect radiant heat (1); Protect from burns (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use a lamp inside a bell jar (1); Sense warmth outside (1); Heat crosses by radiation (1).	1+1+1
16	Stops spread of fire (1); Reduces fuel (1); Controls radiation and convection (1).	1+1+1
17	Hot water rises (1); Convection currents heat all (1); Faster heating (1).	1+1+1
	OR: Day sea breeze cools (1); Night land breeze (1); Moderate temperature (1).	1+1+1
18	Radiation heats (1); Evaporation increases (1); Wind helps (1).	1+1+1
	OR: Radiation passes through glass (1); Seats absorb and heat (1); Hot air trapped (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Land (1). (ii) Sea breeze (1). (iii) Land breeze (1). (iv) At night land cools faster than water (1), so the warm air over the sea rises and the cooler air from the land moves in to take its place (1).	1+1+1+2
20	Radiation reaches the Earth (1); Land heats faster (1); Sea slower (1); Pressure differences (1); Winds blow (1).	1+1+1+1+1
	OR: Air gap between panes (1); Poor conductor (1); Less heat loss (1); Reduces noise (1); Saves energy (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** cork and plastic	Both insulators.
2	**(a)** conduction	Contact.
3	**(a)** feeling warmth from a lit bulb in a closed glass jar	Through vacuum or glass.
4	**(d)** heat it from below	Convection.
5	**(d)** convection currents	Hot rises, cold sinks.
6	**(b)** convection and radiation	Air movement and radiation.
7	**(d)** convection needs a fluid	True.
8	**(c)** radiation	Vacuum cannot stop radiation.
9	**(d)** A is false but R is true.	Light-coloured clothes reflect more radiation and are preferred in summer, so A is false; R is true.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Without a medium, conduction and convection stop, which keeps the liquid hot.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Warm air carries it up (1); Convection (1).	1+1
12	Metal absorbs radiation (1); Conducts heat inside (1).	1+1
13	Reflect sunlight (1); Reduce indoor temperature (1).	1+1
14	Metal conducts heat (1); Wood insulates the hand (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Snow traps air (1); Small entrance retains heat (1); Body heat warms inside (1).	1+1+1
16	No direct radiation (1); Shade reduces heating (1); Less conduction (1).	1+1+1
17	Convection currents (1); Air moves (1); Warm air rises (1).	1+1+1
	OR: Steam raises pressure (1); Higher boiling point (1); Faster cooking (1).	1+1+1
18	Warm air goes out at the top (1); Cold air enters at the bottom (1); Convection currents (1).	1+1+1
	OR: Increase heat loss (1); Moving air cools (1); Blood vessels near the surface (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Radiation (1). (ii) White colour reflects more heat radiation (1). (iii) Conduction and convection (1). (iv) Shiny surfaces reflect heat radiation back (1), so that the tea does not lose heat by radiation and stays hot (1).	1+1+1+2
20	Place a lamp behind glass (1); Feel warmth through glass (1); Glass does not conduct quickly (1); Touching the glass later (1); Different modes (1).	1+1+1+1+1
	OR: Insulate roof (1); Use curtains (1); Close doors (1); Use lids while cooking (1); Solar heaters (1).	1+1+1+1+1

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